Dual Prediction-Guided Distillation for Object Detection in Remote Sensing Images

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***Abstract*— Knowledge distillation, which intends to transfer the expertise from a complex teacher model to a concise student model, has achieved impressive success in object detection. However, many existing distillation methods are designed for object detection tasks on natural images and perform poorly on more challenging remote sensing images. In this study, we first attribute the deficiencies of knowledge distillation in remote sensing object detection to two core reasons: 1) lack of relation distillation among different instances and pixels and 2) significant differences in feature magnitude between the teacher–student pair. Then, we propose a dual prediction-guided knowledge distillation framework, which includes relation distillation and output distillation to address two issues, respectively. Prediction-guided relation distillation (PGRD) is proposed to capture the knowledge of global relation at both instance-wise and pixel-wise, then allowing students to better understand and depict the features distribution across different categories. Prediction-guided output distillation (PGOD) is proposed to mitigate the impact of feature magnitude inconsistencies on distillation with classification and location knowledge, then allowing students to directly capture task-relevant information. Finally, experimental results have indicated the consistent effectiveness of our method across anchor-based two-stage, one-stage, and anchor-free detectors with 11 comparison knowledge distillation methods on three remote sensing detection datasets. Source codes are available at https://github.com/RQ-W/DPGD.git.**

***Index Terms*— Knowledge distillation, nonlocal module, prediction-guided, remote sensing object detection.**

I. Introduction

DEEP learning has witnessed significant advancements in remote sensing detection, offering innovative solutions to practical problems [1], [2], [3], [4], [5], [6], [7]. Generally, many state-of-the-art detectors emphasize precision over inference speed, achieving notable performance despite their large parameter sizes. However, the high demands for memory and computational resources constrain the deployment of these detectors in resource-limited environments such as autonomous vehicles, mobile devices, and satellite on-orbit processing systems. Therefore, there is an urgent need to develop models that trade-off between the complexity and resource efficiency, making this topic a prominent research hotspot.

To address the above-mentioned issue, researchers have proposed various compression techniques, including pruning [8], [9], [10], quantization [11], [12], [13], concise network architecture design [14], [15], [16], and knowledge distillation [17], [18]. Of particular interest is knowledge distillation, also known as teacher–student learning, which offers an elegant way to train a compact student network by leveraging the knowledge distilled from a well-performing teacher network. Initially designed for image classification, vanilla distillation [17] enables the student to effectively inherit the rich knowledge contained in soft label outputs of the teacher. Regrettably, this method often result in only marginal improvements for object detection, primarily due to the additional need for precise localization.

Recently, feature-based distillation on feature pyramid network (FPN) [19] features has been utilized to improve both classification and regression head subtasks. For example, Li et al. [20] concentrated on applying distillation loss to features sampled by the region proposal networks (RPNs). Fine-grained distillation at the feature-level was proposed by Wang et al. [21] selectively transferred feature knowledge related to the near object anchor locations. Nevertheless, the aforementioned methods are tailored for natural image detection and tend to yield marginal effects when employed for remote sensing detection [22]. Here, we attribute the reasons for unsatisfactory distillation on remote sensing detection to two crucial aspects, which will be meticulously analyzed and addressed.

Lack of relation distillation among different instances and pixels. As shown in Fig. 1, instances in remote sensing images commonly exhibit dense arrangements, random orientations, and significant size variations, which are distinct from those in natural images [23]. These characteristics lead to challenges of intraclass diversities and interclass consistencies. Distilling coarse-grained relation knowledge among foreground instances can improve the student's understanding and depiction of feature distribution across different classes [24], as shown in Fig. 2(a). Obviously, it is unwise to consider

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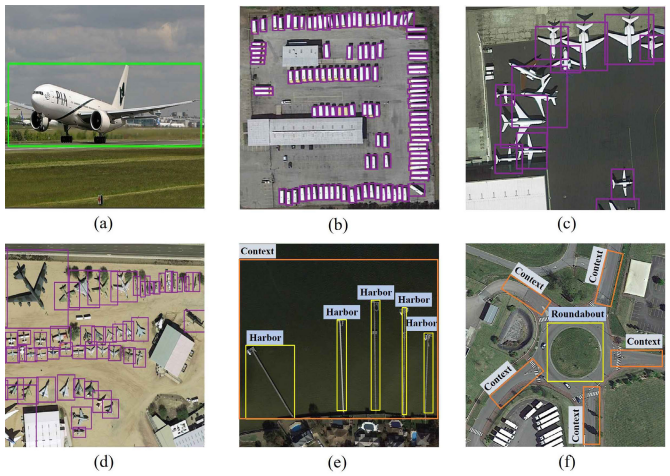


Fig. 1. Visualization of the natural image (PASCAL VOC dataset) and remote sensing images (DOTA dataset). (a) Natural images are captured from a horizontal perspective. (b) Small vehicles exhibit dense arrangements. (c) and (d) Airplanes show variability in both orientation and size. (e) Harbors are generally surrounded by seawater. (f) Roundabouts are typically bordered by radial roads.

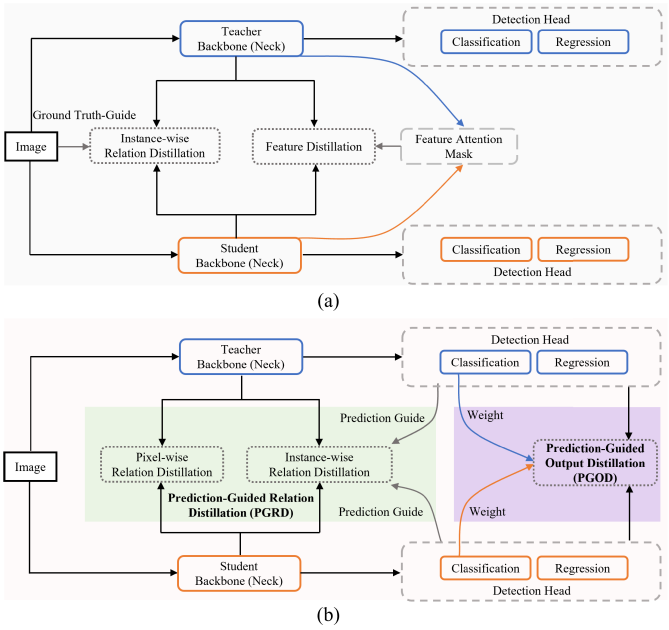


Fig. 2. Different knowledge distillation pipelines for object detection. (a) Existing mainstream methods include ground truth-guided relation distillation and feature-based distillation. (b) We propose PGRD and output distillation.

solely on foreground while disregarding background instances. As shown in Fig. 1(e) and (f), remote sensing objects are typically closely related to their surroundings. Therefore, it is crucial to consider both background and foreground regions. Additionally, the intrinsic pixel relation information covers both spatial relationships and global structures, aiding the model in better understanding pixel representations. Consequently, adopting the fine-grained relation distillation approach becomes essential for accurately discriminating pixels representing distinct classes.

Significant Differences in Feature Magnitude Between the Teacher–Student Pair. In contrast to natural object detection images, the geographic heterogeneity characterizing remote sensing images introduces abundant misleading background perturbations for foreground spatial regions [25]. Thus, the teacher and the student selectively focus on or ignore certain features based on their individual capabilities in such intricate backgrounds, aggravating the differences in feature magnitude. As shown in Fig. 3, we observe that the wide-ranging distribution of feature values in the student contrasts with the concentrated distribution in the teacher, which can lead to information inconsistency. Such inconsistency may cause substantial errors in the distillation objective function and introduce additional noise. Despite the potential benefits of utilizing rich features provided by the teacher for the student, feature-based distillation methods are sensitive to the data variations, resulting in unsatisfactory performance for specific scenarios. Although many mask-based techniques, such as mainstream feature attention masks [see Fig. 2(a)] [21], have been developed to optimize the student's focus, they still yield suboptimal performance.

To address the aforementioned challenges, we propose dual prediction-guided distillation (DPGD) framework, which consists of two parts as illustrated in Fig. 2(b). Prediction-guided relation distillation (PGRD) addresses the existing deficiencies of relation-based distillation for remote sensing detection, which facilitates comprehensive distillation at both instance-wise and pixel-wise[1]. We design a joint prediction-guided dynamic strategy to select foreground instances and valuable discriminative patches from many negatives, effectively utilizing the inner structure of semantic instances. Moreover, previous distillation methods often neglected the interpixel relation [24], [26], [27]. As demonstrated by the success of nonlocal modules [28] and relation networks [29] in enhancing detector performance, investigating pixel-wise relation distillation is essential for complementing instance-wise distillation efforts. Consequently, we use the nonlocal module to capture pixel relations between the student–teacher pair. Subsequently, the relation learning is enhanced by distilling it from the teacher to the student.

On the other hand, feature-based distillation methods might convey considerable task-irrelevant information throughout the entire feature map in more complex remote sensing images. To address this issue, we propose prediction-guided output distillation (PGOD), which directly enables the student to learn the essential feature representations for detection tasks. PGOD includes both classification and location output distillation, facilitating the transfer of diverse knowledge types. Specifically, for classification, we utilize the binary cross-entropy distillation loss to ensure consistency with the original classification task's objectives. For localization, we utilize the most essential spatial relationship between the two predicted bounding boxes, generalized intersection over union (GIoU), as the localization distillation destination. Further inspired by feature masks, we design a joint prediction-guided mask to differentiate critical samples dynamically.

In brief, we propose DPGD framework comprises components based on both relation and output distillation. This method is deployed only during training, neither altering the

[1]In this paper, "pixel-wise" refers to any two pixels in the feature map.

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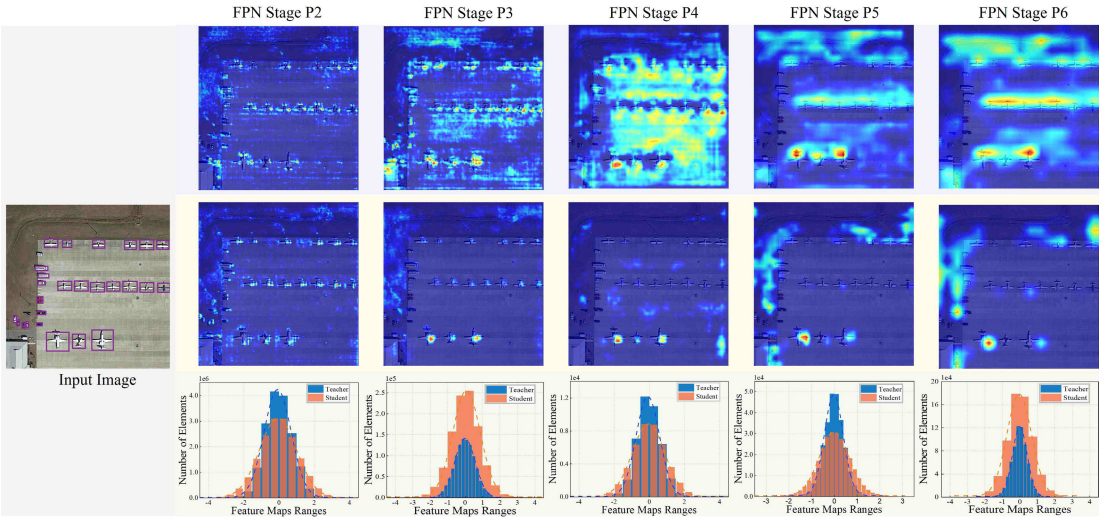


Fig. 3. Example shows the differences in feature magnitude between student and teacher detectors. (Left) Input image. (Right) First and second lines on the display the activation patterns of the Faster RCNN student (ResNet18 backbone) and the Faster RCNN teacher (ResNet101 backbone) at different FPN stages on DOTA dataset, respectively. The third line compares the feature magnitudes between the teacher–student pair.

student architecture nor increasing inference costs. Moreover, entirely agnostic to specific detection structures like RPNs or anchors, DPGD seamlessly integrates into various detectors. Without the bells and whistles, our method can significantly improve the performance of the student detector. In a nutshell, the contributions of this study are as follows.

1) We propose PGRD, which incorporates coarse-to-fine-grained relation distillation. This method not only captures instance-wise relations for both foreground and important background, but also includes pixel-wise relations. It enables the student model to better understand hierarchical structures and contextual relationships in remote sensing detection.
2) We propose PGOD, which combines location-aware with classification information. This method more directly and efficiently conveys crucial patterns for object detection in complex remote sensing environments, outperforming feature-based distillation.
3) We propose a simple yet effective DPGD framework designed to develop precise and compact remote sensing detectors. The effectiveness of our method has been demonstrated through extensive experiments on various datasets, including optical DOTA, DIOR, and SAR, using different detectors (such as anchor-based one-stage, two-stage, and anchor-free models), consistently achieving state-of-the-art performance.

The subsequent structure of this article is organized as follows. Section II provides a review of the relevant literatures. Section III details the proposed method. Section IV presents comprehensive experimental results. Finally, Section V presents a conclusion and illustrates future goals.

II. Related Work

A. Object Detection

Object detection is one of the most challenging tasks in computer vision, primarily focused on locating and classifying regions of interest within images. Scholarly works in this field can be broadly divided into anchor-based and anchor-free detectors. Anchor-based networks can further be divided into two-stage and single-stage detectors. Representative two-stage object detectors like Fast RCNN [30], Faster RCNN [31], and Cascade RCNN [32] conventionally utilize RPNs to generate proposals, subject to refined classification and location adjustment through classifiers and regressors. These networks tend to achieve higher detection accuracy but also increase computational burden. On the other hand, single-stage anchor-based detectors like RetinaNet [33], SSD [34], and the YOLO series [35], [36], [37] simplify the detection process by densely predicting targets directly within a single network framework, thereby eliminating the step of explicit candidate region proposals. This strategy benefits speed, albeit at the expense of some accuracy. Meanwhile, anchor-free detectors remove traditional predefined anchor boxes, further simplifying the detection pipeline. For example, FCOS achieves efficient and accurate detection through pixel-level direct prediction and a "center-ness" scoring mechanism [38]. GFL enhances dense object detection performance by simultaneously optimizing classification and localization accuracy and modeling the positions of bounding boxes more flexibly [39].

The inherent properties of remote sensing images limit the effectiveness of conventional object detection methods designed for natural images, including high-performing two-stage anchor-based detectors. Additionally, the need for real-time operations in remote sensing detectors emphasizes the importance of balancing computational efficiency and performance. In recent years, innovative detection methods specifically for remote sensing have been developed. Wu et al. designed a spatial frequency feature extraction method to tackle rotated bounding boxes in remote sensing images [40]. Cheng et al. [41] designed an contextual attention enhancement module to better enhance the network's ability to learn global information. Wang et al. [42] fused multiscale features by atrous convolution and proposed a region-weighted loss

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function to focus the network on small targets. At the same time, Sun et al. [43] designed a detector without anchor for the garbage dumps detection by enhancing the extracted features by utilizing the attention mechanism. To deal with the misalignment between the target and the anchor in remote sensing detection, Xu et al. [44] used the proposed core anchor instead of the anchor at the regular position. Shamsolmoali et al. [45] coupled an image pyramid with a one-stage detector to enrich small-target features in response to the limited ability of current detectors to perform effective feature extraction on small-scale objects. To better obtain the global semantic knowledge and features of dense tiny objects, Zhao et al. [46] proposed SCDNet, which decouples scene contextual information to recognize more small targets. Currently, one of the primary challenges for detection tasks is addressing the accuracy-speed trade-off [47]. Therefore, we propose DPGD to advance research in this field further.

B. Knowledge Distillation

Knowledge distillation, introduced by Hinton et al. [17], has emerged as an influential technique in model compression, widely acclaimed for its excellent outcomes and operational simplicity in image classification. As research has progressed, knowledge distillation has developed into three principal categories based on the nature of knowledge representation: output-based [48], [49], feature-based [50], [51], and relation-based distillation [52], [53]. While output-based and feature-based distillations concentrate on the relative probabilities of the output layer and the feature representations of the intermediate layer, relation-based distillation captures similarities among data instances.

In the object detection task, the extreme imbalance between the foreground and background presents a notable challenge, complicating the direct application of traditional distillation methods. Recent distillation methods customized for object detection frequently employ mask-based strategies to transfer appropriate foreground features, effectively addressing this challenge selectively. Wang et al. [21] proposed that differences in feature responses near object locations reflected the generalization capability of the teacher detector. Sun et al. [54] introduced the Gaussian mask to cover the ground truth areas for distillation. Zhang and Ma [55] proposed the attention mask on feature maps to facilitate the student's learning. Guo et al. [56] emphasized the equal importance of background areas compared to the foreground and designed different weights for both. Yang et al. [57] utilized attention masks to guide the student in learning essential pixels and channels within foreground and background contexts.

In remote sensing detection, distillation methods also extract crucial feature knowledge to convey rich spatial information. Due to the complexity of remote sensing images, Yang et al. [26] proposed the region-weighted binarization mask strategy that emphasized important foreground areas. Li et al. [24] adopted the unique instance-aware imitation mask generated by evaluating the similarity between class prototypes and pixel embeddings while also considering relation distillation among foreground instances. Shamsolmoali et al. [58] introduced the sophisticated feature mask combining both spatial and channel attention. Yang et al. [27] designed two-way assistant modules that influence both teacher and student features to minimize complex background disturbances. Unlike existing work, we observe that feature-based distillation introduces excessive noise when there is a substantial discrepancy between feature attentions, potentially impacting the distillation effectiveness and leading to suboptimal performance. Furthermore, relying solely on instance-wise relation distillation guided by ground truths to address intraclass variances and interclass similarities is insufficient [24]. There should be a deeper exploration of relation distillation between foreground and background instances, as well as pixel-wise relation distillation, to achieve better performance improvements for detectors.

III. Methodology

We propose DPGD, which enhances the capability of student detector to better learn relation-based and output-based knowledge contributed by teacher detector in remote sensing detection tasks, as illustrated in Fig. 4. This section first provides a brief overview of previous knowledge distillation methods for object detection in Section III-A. Then, Sections III-B and III-C elaborate on PGRD and PGOD, respectively. Finally, Section III-D presents the overall loss function of student training.

A. Preliminaries

In general, given an input image, an object detector first extracts semantic features through a backbone network (such as VGG [59] or ResNet [60]). These features are then fused using a neck layer (such as FPN [19]). Finally, classification logits $\boldsymbol{p'} \in \mathbb{R}^{L\times H\times W\times K}$ and location offsets $\boldsymbol{b'} \in \mathbb{R}^{L\times H\times W\times 4}$ are obtained from the classification and localization heads, respectively. Here, H and W, respectively, denote the height and width of the current feature map, K is the number of foreground classes, and L is the number of FPN layers. The classification scores $\boldsymbol{p} \in \mathbb{R}^{L\times H\times W\times K}$ and bounding boxes $\boldsymbol{b} \in \mathbb{R}^{L\times H\times W\times 4}$ can be computed as

$$\boldsymbol{p} = \text{Sigmoid}\left(\boldsymbol{p'}\right) \tag{1}$$

$$\boldsymbol{b} = \text{Decoder}\left(\boldsymbol{b'}\right) \tag{2}$$

where Decoder$(\cdot)$ denotes the decoding process that transforms bounding box offsets into actual coordinates. For example, anchor-based detectors apply center point coordinate offsets as well as width and height offsets to the anchor boxes to obtain the final bounding box positions. In object detection, the sigmoid operation [i.e., Sigmoid$(\cdot)$] is commonly employed to transform classification logits into scores.

Feature-based distillation has emerged as the predominant method in recent knowledge distillation for remote sensing detection, due to the abundant and diverse features $\boldsymbol{F} \in \mathbb{R}^{L\times H\times W\times C}$ generated by FPN. The objective can be formulated as follows:

$$\mathcal{L}_{\text{fea}} = \sum_{l=1}^{L}\frac{1}{N_f}\sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C}\boldsymbol{M}^f\left(\boldsymbol{F}_l^t - \phi^f\left(\boldsymbol{F}_l^s\right)\right)^2 \tag{3}$$

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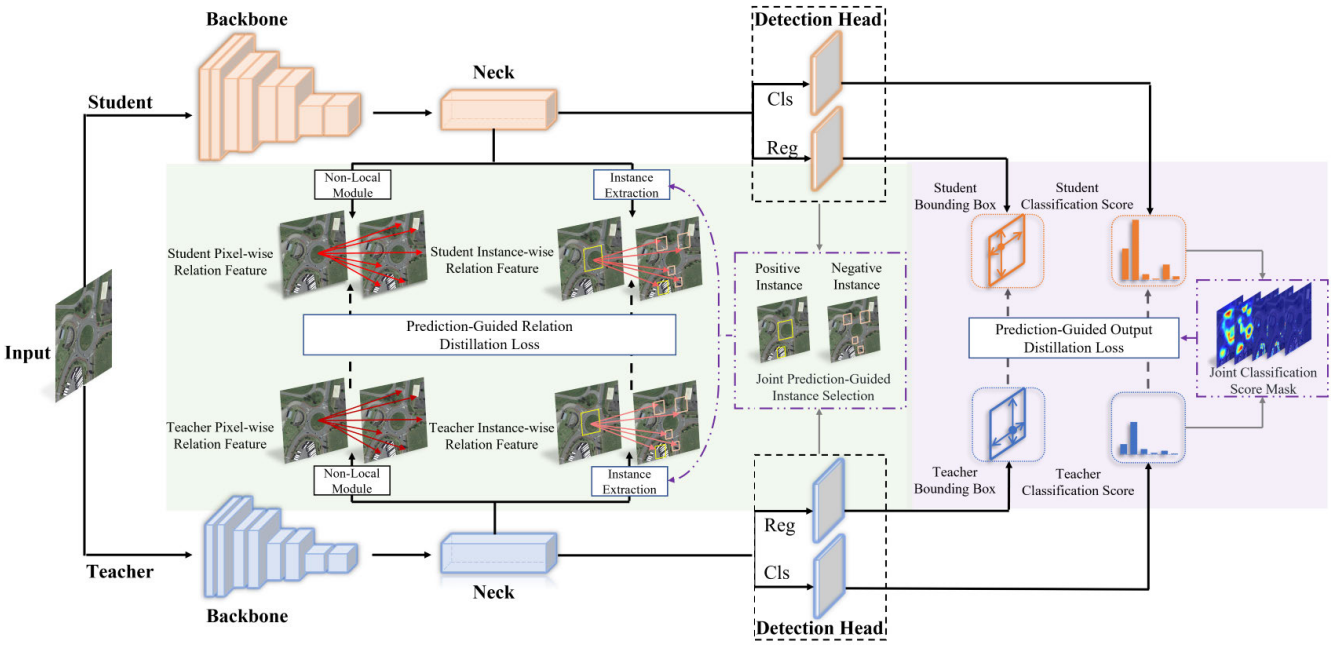


Fig. 4. Details of proposed DPGD method. PGRD and output distillation are merged to better transfer knowledge for remote sensing detection tasks.

where l denotes the lth layer of the FPN features. h, w, and c are denoted as the current feature map's coordinate position and channel number with height H, width W, and channel number C, respectively. $N_f = \sum_{h=1}^{H}\sum_{w=1}^{W}\sum_{c=1}^{C} \boldsymbol{M}^f$ is a normalization factor. $\boldsymbol{F}_l^t$ and $\boldsymbol{F}_l^s$ are the lth layer FPN features of the teacher and student detectors, respectively. $\phi^f(\cdot)$ is a linear transformation layer, usually a 1×1 convolution layer, used to align the feature dimensions of the student and teacher detectors. Mask $\boldsymbol{M}^f$ is defined differently in various methods, and it helps focus on the foreground while filtering out background noise.

Regrettably, existing methods overlook the significant attention gaps between features caused by the complex environment in remote sensing, as illustrated in Fig. 3. Even though different sophisticated masks have been proposed to direct the student model's attention toward valuable regions, these methods still lead to suboptimal performance (refer to Section IV-D). We demonstrate that the DPGD framework, using only relation-based and output-based distillation, enables the student model to learn task-specific knowledge directly for object detection. This method avoids the extensive noise introduced in feature distillation process, significantly improving student performance.

B. Prediction-Guided Relation Distillation

PGRD aims to assist the student in effectively distinguishing challenging remote sensing instances through comprehensive relation distillation at both instance-wise and pixel-wise. Following conventional descriptions, we will elaborate on these methods in detail.

1) Instance-Wise Relation Distillation: Previous distillation methods designed ground truth-guided instance-wise relation distillation to mitigate the severe intraclass variances and interclass similarities in remote sensing detection. Notably, remote sensing instances often demonstrate strong associations with their backgrounds. As shown in Fig. 1(e) and (f), harbors are predominantly surrounded by seawater, and roundabouts are typically encircled by roads. Extracting instance relations solely from ground truth regions may inadvertently overlook critical background details [56], [57]. Therefore, adopting a more adaptive mechanism for dynamically selecting instances is imperative rather than relying exclusively on static ground truth-guided.

To this end, we propose a joint prediction-guided dynamic strategy, as illustrated in Fig. 5, carefully selecting foreground and informative background instances from many negatives.

a) Joint prediction-guided instance selection: Let $\boldsymbol{p}^t \in \mathbb{R}^{L\times H\times W\times K}$ and $\boldsymbol{p}^s \in \mathbb{R}^{L\times H\times W\times K}$ represent the classification scores of the teacher and the student detectors, respectively. $\boldsymbol{b}^t \in \mathbb{R}^{L\times H\times W\times 4}$ and $\boldsymbol{b}^s \in \mathbb{R}^{L\times H\times W\times 4}$ represent the bounding boxes of the teacher and student detectors, respectively. Predicted scores offer a measure of confidence in the objects and the informativeness of the regions [61]. Therefore, dynamically selecting optimal instances based on the joint prediction of both models is a reasonable and straightforward approach. Initially, as depicted in Fig. 5(a), we obtain candidate instances matching the number of predicted instances, with their classification scores $\boldsymbol{p}^{\mathrm{CI}}$ and bounding boxes $\boldsymbol{b}^{\mathrm{CI}}$ are defined as

$$\boldsymbol{p}_r^{\mathrm{CI}} = \max\left(\max_{0<k\le K} \boldsymbol{p}_{r,k}^t, \max_{0<k\le K} \boldsymbol{p}_{r,k}^s\right) \tag{4}$$

$$\boldsymbol{b}_r^{\mathrm{CI}} = \begin{cases} \boldsymbol{b}_r^t, & \max\limits_{0<k\le K} \boldsymbol{p}_{r,k}^t > \max\limits_{0<k\le K} \boldsymbol{p}_{r,k}^s \\ \boldsymbol{b}_r^s, & \max\limits_{0<k\le K} \boldsymbol{p}_{r,k}^t \le \max\limits_{0<k\le K} \boldsymbol{p}_{r,k}^s \end{cases} \tag{5}$$

where $R = L \times H \times W$ and K denote the number of predicted bounding boxes and classes, respectively. r and k are indexes of dimensions R and K, subject to $0 < r \le R$ and $0 < k \le K$. $\boldsymbol{p}_{r,k}^t$ and $\boldsymbol{p}_{r,k}^s$ denote the kth class of the rth

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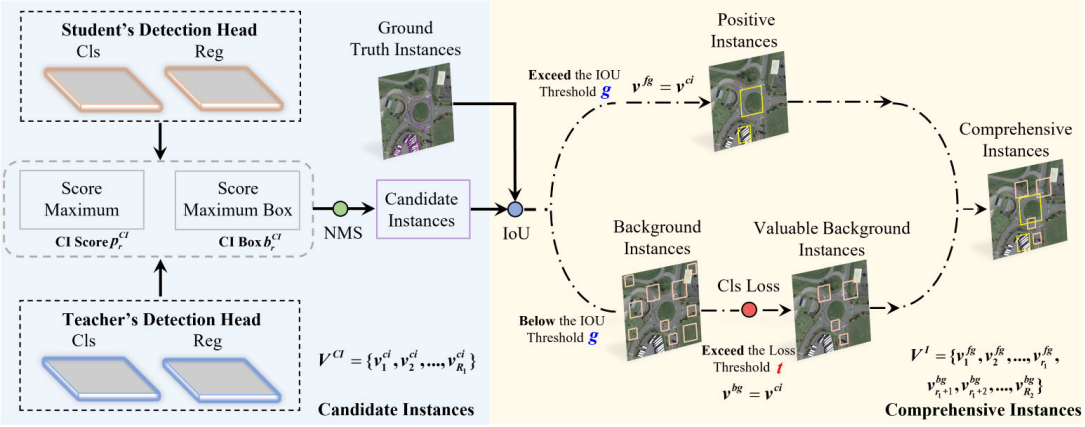


Fig. 5. Details of our joint prediction-guided instance selection method. (a) We select the higher of predicted classification scores from the teacher and the student as the candidate instances scores, and bounding boxes corresponding to these scores as the candidate instances boxes. (b) We calculate the IoU values between candidate instances and ground truths, and assign those with higher IoU values as positive instances and the rest as negatives. Additionally, negative instances with elevated classification losses are prioritized as valuable patches.

prediction instance of the teacher and student, respectively. For anchor-based models like Faster RCNN [31], $\boldsymbol{p_r}$ represents the score that the object is a foreground. For anchor-free models like FCOS [38], $\boldsymbol{p_r}$ represents scores for various classes. $\boldsymbol{b_r^t}$ and $\boldsymbol{b_r^s}$ denote the rth predicted teacher and student bounding box, respectively. For the rth candidate instance, the classification score $\boldsymbol{p_r^{\mathrm{CI}}}$ is obtained by taking the higher value between the teacher and the student classification scores using the max$(\cdot)$ operation. Its corresponding box is used as the candidate instance's box $\boldsymbol{b_r^{\mathrm{CI}}}$.

Inspired by the region proposals in anchor-based detectors like Fast R-CNN [30], we meticulously select valuable instances from the candidate instances. Unlike those methods, ours does not involve components such as anchors or RPNs, simplifying integration into various detectors. Specifically, since the vast number of candidate instances, we apply non-maximum suppression (NMS) with the threshold u to avoid duplicated computations, which can be formulated as

$$\begin{aligned} V^{\mathrm{CI}} &= \mathrm{NMS}\left(\boldsymbol{p}^{\mathrm{CI}}, \boldsymbol{b}^{\mathrm{CI}}; u\right), \quad \text{and} \\ V^{\mathrm{CI}} &= \left\{v_1^{\mathrm{ci}}, v_2^{\mathrm{ci}}, \ldots, v_{R_1}^{\mathrm{ci}}\right\} \end{aligned} \tag{6}$$

where V^{CI} is the set of candidate instances after the NMS operation, and R_1 is its total number. NMS iteratively selects high-score candidate instances and discards lower score instances with substantial overlaps. Subsequently, as depicted in Fig. 5(b), the maximum intersection over union (IoU) values between these candidate instance boxes and each ground truth box are calculated. Set instances with IoU values exceeding the threshold g as valuable foreground instances, while others as background instances, which can be formulated as

$$\begin{aligned} v^{\mathrm{fg}} &= v^{\mathrm{ci}}, \quad \text{s.t.} \max_{0<r'\leq R'} \left(\mathrm{IoU}\left(\boldsymbol{b}^{\mathrm{ci}}, b_{r'}^{\mathrm{gt}}\right)\right) > g, \ \text{and} \\ V^{\mathrm{CI}} &= \left\{v_1^{\mathrm{fg}}, v_2^{\mathrm{fg}}, \ldots, v_{r_1}^{\mathrm{fg}}, v_{r_1+1}^{\mathrm{ci}}, v_{r_1+2}^{\mathrm{ci}}, \ldots, v_{R_1}^{\mathrm{ci}}\right\} \end{aligned} \tag{7}$$

where $\boldsymbol{b}^{\mathrm{ci}}$ denotes the bounding box of candidate instance v^{ci}. $b_{r'}^{\mathrm{gt}}$ denotes the bounding box of r'th ground truth, subject to $0 < r' \leq R'$. v^{fg} denotes the foreground instance, and r_1 denotes its total number. Note that r_1 dynamically changes as the student training.

In general, most of the candidate instances V^{CI} are negative instances (i.e., $r_1 << R_1 - r_1$), which can overwhelm the contribution of foreground instances and pose challenges to the training process. However, pruning all background instances would result in the loss of critical information, leading to an insufficient understanding of the entire scenario by the detector. Therefore, inspired by OHEM [62], we identify qualified background instances v^{bg} in v^{ci} with the maximum binary cross-entropy loss greater than the threshold t, which can be formulated as

$$\mathcal{L}_{\mathrm{BCE}}\left(\boldsymbol{p_k^{\mathrm{ci}}}, p_k^{\mathrm{gt}}\right) = -\left(\left(1 - p_k^{\mathrm{gt}}\right) \cdot \log\left(1 - \boldsymbol{p_k^{\mathrm{ci}}}\right) + p_k^{\mathrm{gt}} \cdot \log\left(\boldsymbol{p_k^{\mathrm{ci}}}\right)\right) \tag{8}$$

$$v^{\mathrm{bg}} = v^{\mathrm{ci}}, \quad \text{s.t.} \max_{0<r'\leq R'} \left(\sum_{k=1}^{K} \mathcal{L}_{\mathrm{BCE}}\left(\boldsymbol{p_k^{\mathrm{ci}}}, p_{r',k}^{\mathrm{gt}}\right)\right) > t \tag{9}$$

where $\boldsymbol{p_k^{\mathrm{ci}}}$ denotes the kth class of predicted scores for candidate instances v^{ci}, and $p_{r',k}^{\mathrm{gt}}$ denotes the kth class of predicted scores for the r'th ground truth. $\mathcal{L}_{\mathrm{BCE}}(\cdot)$ stands for binary cross-entropy loss. These instances with large classification losses are intrinsically prone to misclassification, hence they are reasonably selected as valuable. Finally, we obtain the set of selected instances V^I as follows:

$$V^I = \left\{v_1^{\mathrm{fg}}, v_2^{\mathrm{fg}}, \ldots, v_{r_1}^{\mathrm{fg}}, v_{r_1+1}^{\mathrm{bg}}, v_{r_1+2}^{\mathrm{bg}}, \ldots, v_{R_2}^{\mathrm{bg}}\right\} \tag{10}$$

where R_2 is the total number of selected instances V^I.

b) Instance feature extraction: Unlike previous methods that map instances to a unified feature layer (e.g., C5 features in a ResNet backbone [24]), we map the selected instances to different layers of multiscale FPN features based on their sizes, which is consistent with the mapping mechanism of the original FPN [19], which can be formulated as

$$\boldsymbol{M_{h,w}^i} = \begin{cases} 1, & \text{if } (h, w) \in \boldsymbol{b^i} \\ 0, & \text{Otherwise} \end{cases} \tag{11}$$

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$$\boldsymbol{f}^{\boldsymbol{i}} = \xi\left(\boldsymbol{M}^{\boldsymbol{i}} \boldsymbol{F}_{\boldsymbol{l}}^{\boldsymbol{i}}\right) \tag{12}$$

where $\boldsymbol{b}^{\boldsymbol{i}}$ denotes the bounding box of the ith selected instance V^I. h and w are the coordinates on the feature map. $\boldsymbol{M}^{\boldsymbol{i}} \in \{0,1\}^{H\times W}$ is the binary mask for the ith selected instance, which takes the value of 1 when (h, w) falls within the selected instance's region; otherwise, it is 0. $\boldsymbol{F}_{\boldsymbol{l}}^{\boldsymbol{i}} \in \mathbb{R}^{H\times W\times C}$ denotes the lth layer FPN feature corresponding to the selected instance. $\xi(\cdot)$ represents the ROI Align operation [2], which adjusts the instance feature to a uniform size $\boldsymbol{f}^{\boldsymbol{i}} \in \mathbb{R}^{O\times O\times C}$ (usually $7 \times 7 \times C$). Based on the selected instance set V^I, we can obtain all instance features $\boldsymbol{f}^{\boldsymbol{I}} \in \mathbb{R}^{R_2\times O\times O\times C}$ and flatten them for subsequent computations.

c) Instance-wise relation distillation loss: Let $\boldsymbol{f}_{\boldsymbol{p}}^{\boldsymbol{I}}$ and $\boldsymbol{f}_{\boldsymbol{q}}^{\boldsymbol{I}}$ denote vector representations in the R_2 selected instance. The relation between instances can be characterized by their similarity in the embedding space. We use cosine similarity to measure this similarity $\boldsymbol{s}_{\boldsymbol{p},\boldsymbol{q}}{}^{\boldsymbol{I}}$ as follows:

$$\boldsymbol{s}_{\boldsymbol{p},\boldsymbol{q}}{}^{\boldsymbol{I}} = \frac{\boldsymbol{f}_{\boldsymbol{p}}^{\boldsymbol{I}} \cdot \boldsymbol{f}_{\boldsymbol{q}}^{\boldsymbol{I}}}{\left\|\boldsymbol{f}_{\boldsymbol{p}}^{\boldsymbol{I}}\right\|_2 \cdot \left\|\boldsymbol{f}_{\boldsymbol{q}}^{\boldsymbol{I}}\right\|_2}. \tag{13}$$

We can, respectively, extract the selected instance feature representations $\boldsymbol{f}^{\boldsymbol{t},\boldsymbol{I}} \in \mathbb{R}^{R_2\times OOC}$ and $\boldsymbol{f}^{\boldsymbol{s},I} \in \mathbb{R}^{R_2\times OOC}$ from the teacher and the student, and calculate the overall relation similarity $\boldsymbol{s}^{\boldsymbol{t},I} \in \mathbb{R}^{R_2\times R_2}$ and $\boldsymbol{s}^{\boldsymbol{s},I} \in \mathbb{R}^{R_2\times R_2}$ according to (13). Consequently, the instance-wise distillation loss is defined as follows:

$$\mathcal{L}_{\text{ins}} = \frac{1}{{R_2}^2} \sum_{r_2=1}^{R_2} \sum_{r_2=1}^{R_2} \left\| \boldsymbol{s}_{\boldsymbol{i},\boldsymbol{j}}^{\boldsymbol{t},\boldsymbol{I}} - \boldsymbol{s}_{\boldsymbol{i},\boldsymbol{j}}^{\boldsymbol{s},\boldsymbol{I}} \right\|^2. \tag{14}$$

We utilize the mean squared error loss to convey structured knowledge based on global instance relations, enabling the student to learn more useful information.

2) Pixel-Wise Relation Distillation: The aforementioned instance-wise relation distillation conveys valuable interregion relation information, which can be considered coarse-grained relation distillation. Additionally, we consider fine-grained pixel-wise relations. The nonlocal module [28] effectively improves model performance by capturing global pixel relation information. It has also been widely used in remote sensing detection [41], [63]. Unlike previous nonlocal networks that use this module to enhance features, we apply the nonlocal module only during the student training phase to extract relations between pixels (see Fig. 6), which can be formulated as

$$y_{i,j} = \frac{1}{N_n} \sum_{i'=1}^{H} \sum_{j'=1}^{W} h\left(\boldsymbol{F}_{\cdot,\boldsymbol{i},\boldsymbol{j}}, \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}\right) g\left(\boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}\right) \tag{15}$$

where y denotes the relation representation between any two pixels in the feature map $\boldsymbol{F}_{\cdot,\boldsymbol{i},\boldsymbol{j}}$ are the spatial indexes where responses are to be computed. At the same time, i', j' are spatial indexes of all possible locations in the feature map. N_n is the normalization factor. $h(\cdot)$ computes the relation between pixels, which is represented in this study as computing Gaussian distances in embedding space. $g(\cdot)$ computes the individual pixel feature representation, specified as a 1×1 convolution. They can be formulated as follows:

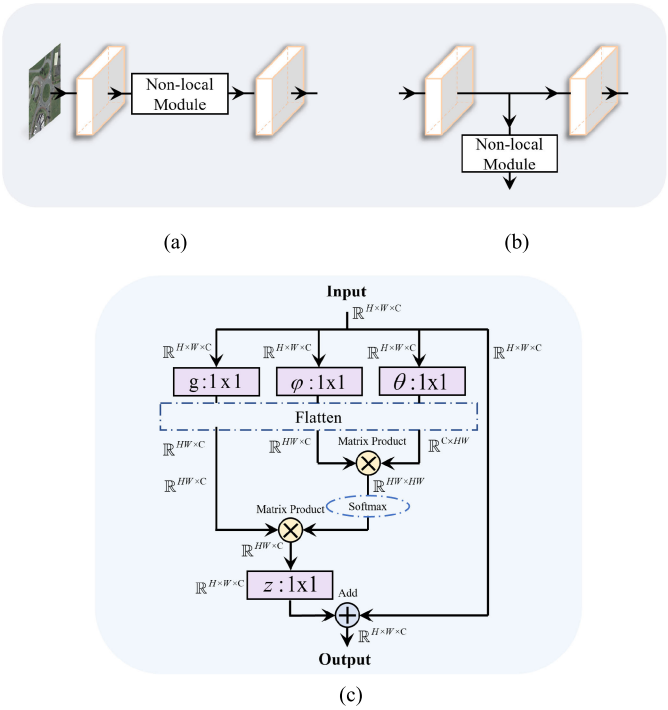


Fig. 6. Details of nonlocal network, nonlocal distillation, and nonlocal module. (a) Nonlocal network. (b) Nonlocal distillation. (c) Nonlocal module.

$$h\left(\boldsymbol{F}_{\cdot,\boldsymbol{i},\boldsymbol{j}}, \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}\right) = e^{\left(W_\theta \boldsymbol{F}_{\cdot,\boldsymbol{i},\boldsymbol{j}}\right)^{\mathrm{T}} W_\varphi \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}} \tag{16}$$

$$g\left(\boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}\right) = W_g \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'} \tag{17}$$

where W_g, W_θ, and W_φ are the weights of different 1×1 convolution kernels. Considering the normalization factor N_n in (15), it can be viewed as the softmax operation. Thus, (15) can be rewritten as follows:

$$y_{i,j} = \sum_{i'=1}^{H} \sum_{j'=1}^{W} \operatorname{softmax}\left(W_\theta^{\mathrm{T}} \boldsymbol{F}_{\cdot,\boldsymbol{i},\boldsymbol{j}}{}^{\mathrm{T}} W_\varphi \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}\right) W_g \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'}. \tag{18}$$

Finally, by adding a residual-like operation [60] to retain part of the original input features, the output of the entire nonlocal module can be more accurately formulated as

$$\boldsymbol{z}_{\boldsymbol{i},\boldsymbol{j}} = W_z y_{i,j} + \boldsymbol{F}_{\cdot,\boldsymbol{i}',\boldsymbol{j}'} \tag{19}$$

where W_z is also a learnable weight of the 1×1 convolution kernel. In both the teacher and student detectors, pixel-wise relation-aware features $\boldsymbol{z}^{\boldsymbol{t}}$ and $\boldsymbol{z}^{\boldsymbol{s}}$ can be obtained through the nonlocal modules, and we use L_2 distance for knowledge transfer

$$\mathcal{L}_{\text{pr}} = \frac{1}{L} \sum_{l=1}^{L} \left\| \boldsymbol{z}_{\boldsymbol{l}}^{\boldsymbol{t}} - \phi_l\left(\boldsymbol{z}_{\boldsymbol{l}}^{\boldsymbol{s}}\right) \right\|_2 \tag{20}$$

where $\boldsymbol{z}_{\boldsymbol{l}}^{\boldsymbol{t}}$ and $\boldsymbol{z}_{\boldsymbol{l}}^{\boldsymbol{s}}$ represent the pixel relations obtained on the lth layer FPN feature. $\phi_l(\cdot)$ is an adaptive convolution layer used to adjust the feature dimensions. L is the number of FPN layers.

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C. Prediction-Guided Output Distillation

Existing remote sensing detection distillation methods often consider feature-based distillation as indispensable components. This is mainly due to its ability to fully utilize the rich semantic information of multiscale FPN features. However, the complex backgrounds of remote sensing images increase the discrepancy in feature attention between teacher and student, often leading to additional noise in feature-based distillation. Even mainstream feature distillation methods based on attention masks frequently achieve suboptimal performance. Therefore, we propose PGOD to directly assist the student detector in learning the most critical information related to detection tasks, which includes classification and localization output distillation. Following conventional descriptions, we will elaborate on these methods in detail.

Furthermore, inspired by mask feature distillation [55], we propose a novel weighting strategy to enhance the student model's focus on critical samples. Specifically, the classification-based method is particularly suitable for achieving this goal, as it directly reflects the probability that a sample belongs to the foreground region. Based on this, we analyze the two different masks based on classification scores.

1) Teacher-Guided Mask: We use the maximum classification scores from the teacher model for each sample as the mask $\boldsymbol{M}^{t} \in \mathbb{R}^{R}$, employed in methods like FRS [64] and two-way [27], defined as

$$\boldsymbol{M}^{t} = \max_{0<k\leq K} \left| \boldsymbol{p}_{k}^{t} \right| \tag{21}$$

where $\boldsymbol{p}_{k}^{t}$ denotes the teacher's predicted score for the kth class. However, it should be noted that such masks are static and do not adjust according to the various training stages. This may result in meaningless samples being weighted while important samples are undervalued for the student.

2) Joint Prediction-Guided Mask: We simply compute the L_1 distance between $\boldsymbol{p}_{k}^{t}$ and $\boldsymbol{p}_{k}^{s}$ as the dynamic weight $\boldsymbol{M}^{t,s} \in \mathbb{R}^{R}$. This indicates that a greater prediction difference for the same sample signifies more information, rendering it more deserving of the student's attention, which can be formulated as

$$\boldsymbol{M}^{t,s} = \max_{0<k\leq K} \left| \boldsymbol{p}_{k}^{t} - \boldsymbol{p}_{k}^{s} \right|. \tag{22}$$

Unlike the static mask generated solely based on the teacher's classifications, this dynamic mask can gradually adjust as the student learns, improving performance more effectively.

Fig. 7 presents a visualization of prediction-guided and attention-guided masks of the Faster R-CNN teacher (ResNet101 backbone) and student (ResNet18 backbone) trained on DOTA. It is observed that prediction-guided static masks consistently focus on global information of the objects while neglecting edge details. Dynamic masks, however, gradually shift their focus toward edge information as training progresses. This shift occurs because edge information effectively captures local features and details of objects, aiding in distinguishing between different object categories [65], [66], [67]. Additionally, it is noted that attention-guided masks [55], [58] are insensitive to small objects and prone to introduce object-unrelated noise, possibly because features with substantial differences can implicitly affect these masks. Therefore, we use the joint prediction-guided dynamic mask to direct the student to focus on samples with larger prediction differences from the teacher, providing a more reasonable learning mechanism.

a) Classification output distillation: In current object detection architectures, different detectors typically use their specific classification loss to optimize performance. For example, RetinaNet [33] uses focal loss, while GFL [39] uses quality focal loss. It is worth noting that these losses evolved from the binary cross-entropy loss, retaining the same core concept to some extent. Consequently, we use the simple binary cross-entropy loss to implement knowledge transfer, consistent with the classification task's core processing. By this means, we aim to improve the efficiency of knowledge distillation while minimizing potential inconsistencies caused by differences in loss functions, thereby achieving a more robust knowledge transfer process, which can be formulated as

$$\mathcal{L}_{\text{cls}} = \frac{1}{N_c} \sum_{r=1}^{R} \boldsymbol{M}_{r}^{t,s} \sum_{k=1}^{K} \mathcal{L}_{\text{BCE}}\left(\boldsymbol{p}_{r,k}{}^{s}, \boldsymbol{p}_{r,k}^{t} \right) \tag{23}$$

where $\mathcal{L}_{\text{cls}}$ denotes the classification distillation loss, and $\mathcal{L}_{\text{BCE}}(\cdot)$ denotes the binary cross entropy loss. $\boldsymbol{M}_{r}^{t,s}$ is the weight value for the rth predicted sample, and $N_c = \sum_{r=1}^{R} \boldsymbol{M}_{r}^{t,s}$ is the normalization factor. $\boldsymbol{p}_{r,k}{}^{s}$ and $\boldsymbol{p}_{r,k}^{t}$ signify the predicted scores for the kth class at the rth position in the student and teacher detectors, respectively.

b) Localization output distillation: Precise localization information is crucial alongside classification in object detection. Different detectors employ various regression methods, such as regressing from predefined anchors to the target's bounding boxes in the anchor-based detector or regressing from specific point locations in anchor-free detectors. To address inconsistencies introduced by disparate regression methods in distillation, we adopt the most fundamental spatial relationship measure between two bounding boxes, IoU, as the distillation objective. Furthermore, we consider using an improved version of IoU loss, the GIoU loss function, as the localization output distillation loss. Compared to the IoU, GIoU loss not only considers the position and size of bounding boxes but also their shapes, thereby offering better stability and constraint, which can be formulated as

$$\mathcal{L}_{\text{GIoU}}\left(\boldsymbol{b}_{r}^{s}, \boldsymbol{b}_{r}^{t} \right) = 1 - \frac{\text{I}\left(\boldsymbol{b}_{r}^{t}, \boldsymbol{b}_{r}^{s} \right)}{\text{U}\left(\boldsymbol{b}_{r}^{t}, \boldsymbol{b}_{r}^{s} \right)} + \frac{\text{A}\left(\boldsymbol{b}_{r}^{t}, \boldsymbol{b}_{r}^{s} \right) - \text{U}\left(\boldsymbol{b}_{r}^{t}, \boldsymbol{b}_{r}^{s} \right)}{\text{A}\left(\boldsymbol{b}_{r}^{t}, \boldsymbol{b}_{r}^{s} \right)} \tag{24}$$

$$\mathcal{L}_{\text{loc}} = \frac{1}{N_l} \sum_{r=1}^{R} \boldsymbol{M}_{r}^{t,s} C\left(\boldsymbol{b}_{r}^{s} \right) \mathcal{L}_{\text{GIoU}}\left(\boldsymbol{b}_{r}^{s}, \boldsymbol{b}_{r}^{t} \right) \tag{25}$$

where $\mathcal{L}_{\text{GIoU}}(\cdot)$ denotes the GIoU loss. $N_l = \sum_{r=1}^{R} \boldsymbol{M}_{r}^{t,s} C(\boldsymbol{b}_{r}^{s})$ denotes the normalization factor. $\boldsymbol{b}_{r}^{t}$ and $\boldsymbol{b}_{r}^{s}$ denote the rth bounding box from the student and teacher detectors, respectively. $\text{I}(\cdot)$ and $\text{U}(\cdot)$ denote the intersection and union areas of the two bounding boxes, respectively. $\text{A}(\cdot)$ denotes the size of the smallest enclosing box of the two rectangles. In object detection, regressing predicted bounding boxes far

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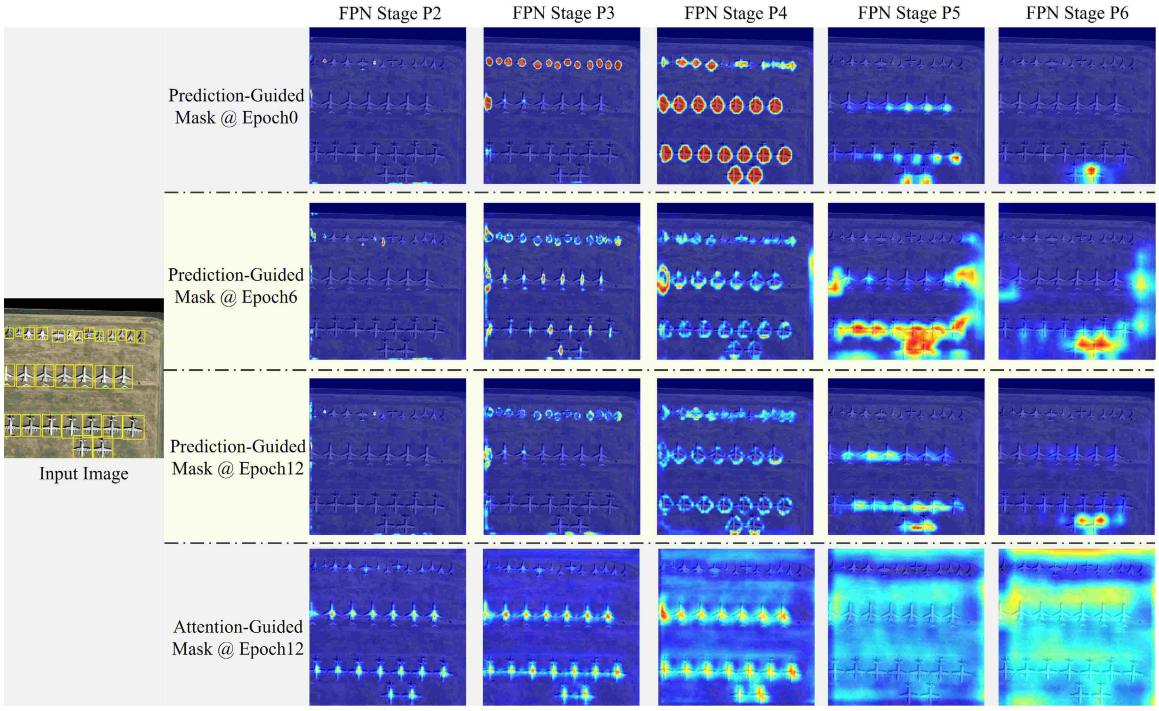


Fig. 7. Example illustrates the differences between prediction-guided masks and attention-guided masks. Note that "prediction-guided mask at epoch0" indicates the teacher-guided static mask (21) of the Faster RCNN student (ResNet18 backbones) on the DOTA dataset. Note that "prediction-guided mask at epoch6" and "prediction-guided mask at epoch12" indicates dynamic masks (22) of the student trained for six epochs and 12 epochs, respectively. "Attention-guided mask at epoch12" indicate the mainstream attention-based mask [55], [58] of the student trained for 12 epochs.

from the ground truth boxes generally does not improve detector performance. Therefore, we introduced the region selection criterion $C(\cdot)$ to distill only the predictions of positive samples. Specifically, we compute the IoU values between the predicted bounding box from the student and all ground truth boxes, selecting the maximum IoU value. If this value exceeds the threshold set g (defined in Section III-B), the predicted sample is considered positive (labeled as 1); otherwise, it is considered negative (labeled as 0).

D. Overall Loss

As previously described, the overall loss function consists of five components. $\mathcal{L}_{\text{det}}$ is the original task loss used to train the student. $\mathcal{L}_{\text{ins}}$ and $\mathcal{L}_{\text{pr}}$ are the global instance-wise relation distillation loss and the pixel-wise relation distillation loss in PGRD, respectively. $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{loc}}$ are the classification output distillation loss and the localization output distillation loss in PGOD. The overall loss function can be formulated as follows:

$$\mathcal{L}_{\text{student}} = \mathcal{L}_{\text{det}} + \underbrace{\alpha\mathcal{L}_{\text{ins}} + \beta\mathcal{L}_{\text{pr}}}_{\text{PGRD}} + \underbrace{\gamma\mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{loc}}}_{\text{PGOD}}, \tag{26}$$

where α, β, γ, and λ are hyperparameters used to balance the different losses in our method. The overall distillation loss is computed only on feature maps and output heads without involving specific components within the detector. Thus, it can be easily applied to various detector architectures.

IV. Experiment

In this section, we describe in detail the datasets used, the evaluation metrics commonly used in object detection, the parameter settings selected, and extensive experiments that demonstrate the effectiveness of our approach. In addition, we present an ablation to further illustrate the contribution of each distillation module within the overall DPGD framework.

A. Datasets

1) DOTA: The DOTA dataset [68] consists of 2806 images containing 188 282 remote sensing instances across 15 categories. Image sizes vary from 800 × 800 to 4000 × 4000, and a single image may include multiple targets with diverse categories, sizes, and orientations. We use half of the original dataset as the training set, 1/6 as the validation set, and 1/3 as the test set. To enhance training efficiency, we crop these images to 1024 × 1024 with an overlap of 200 pixels, yielding 15 749 training patches and 5297 validation patches. We use the training patches for training and report experimental results on the validation patches.

2) DIOR: The DIOR dataset, a benchmark for optical remote sensing object detection [25], consists of 23 463 remote sensing images incorporating 190 288 instances across 20 categories. All images are sized at 800 × 800. For our experiments, we randomly selected 11 725 images for the training set and 11 738 images for the validation set.

3) SSDD: We assess the proposed method on the SSDD [69], analogous to two-way [27]. This is the first publicly available SAR dataset for ship object detection, containing 1160 images and 2456 ships. All images are sized at 600 × 600. For our experiments, we randomly selected 928 images for the training set and 232 images for the validation set.

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B. Evaluation Metrics

We adopt official performance metrics [70] to evaluate the DPGD performance, including average precision (AP), AP50, AP75, AP_S, AP_M, and AP_L. AP involves true positives (TP), false positives (FP), false negatives (FN), precision, and recall. In object detection, a prediction is labeled as TP if the IoU value between its bounding box and the ground truth box exceeds a predefined threshold. If the IoU value is lower than this threshold, the prediction is classified as FP. FN indicates that the prediction result fails to detect an actual object. Accordingly, precision and recall can be calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad \text{and}$$
$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{27}$$

We can obtain different precision and recall values using the confidence scores of various predictions, thereby constructing precision–recall (P–R) curves. The mean AP (mAP) is obtained by calculating the average area under the P–R curve for all classes at a certain IoU threshold. AP denotes the average mAP across different IoU thresholds (from 0.5 to 0.95 with a step size of 0.05).

In addition, we utilize frames per second (FPS) and model parameters to measure the efficiency of the detectors. FPS denotes the number of image frames the detector processes per second during inference, directly reflecting the processing speed. The reported FPS values are measured on an RTX 3090 GPU. The model parameters directly influence the complexity and resource consumption of detectors.

TABLE I

PERFORMANCE OF TEACHER AND STUDENT PAIRS USED IN EXPERIMENTS. FOR ALL THE DETECTION FRAMEWORKS USED, RESNET101 IS THE TEACHER MODEL AND RESNET18 IS THE STUDENT MODEL, WHILE SWIN-BASE IS THE TEACHER MODEL AND SWIN-TINY IS THE STUDENT MODEL

Dataset	Model	Backbone	FPS (img/s)	Params (Mb)	AP (%)
DOTA	Faster RCNN	ResNet101	16.5	60.19	41.9
		ResNet18	32.5	28.19	35.4
		Swin-base	9.9	104.17	43.2
		Swin-tiny	18.4	44.82	41.2
	RetinaNet	ResNet101	17.1	55.39	37.1
		ResNet18	33.9	19.90	32.4
	FCOS	ResNet101	18.5	50.81	37.2
		ResNet18	34.6	18.97	33.0
DIOR	Faster RCNN	ResNet101	25.8	60.21	49.8
		ResNet18	45.0	28.22	41.3
		Swin-base	14.9	104.19	54.6
		Swin-tiny	23.5	44.84	49.1
	RetinaNet	ResNet101	23.3	55.49	45.9
		ResNet18	40.3	20.01	37.1
	FCOS	ResNet101	24.5	50.92	46.6
		ResNet18	43.2	19.07	38.6
SSDD	Faster RCNN	ResNet101	33.7	60.11	59.0
		ResNet18	51.6	28.12	54.0
		Swin-base	25.3	104.09	59.5
		Swin-tiny	35.9	44.75	55.6
	RetinaNet	ResNet101	39.2	55.10	53.0
		ResNet18	62.9	19.61	48.1
	FCOS	ResNet101	39.8	50.88	52.1
		ResNet18	65.0	18.94	47.5

C. Parameter Settings

Our experiments are conducted with various detector architectures, including the anchor-based two-stage Faster R-CNN [31] and one-stage RetinaNet [33], along with the anchor-free FCOS [38]. For these architectures, we employ ResNet-101 for the teacher backbone and ResNet-18 for the student backbone. Furthermore, we provide the teacher–student pair based on Swin Transformer backbone networks [72]. Table I details the original accuracy, parameters, and FPS of both teacher and student detectors. For comparative analysis, we consider 11 existing knowledge distillation methods: KD [17] and AT [71] for classification tasks; FGFI [21], FKD [55], FRS [64], defeat [56], and FGD [57] for object detection tasks; and ARSD [26], InsDist [24], AFD [58], and two-way [27] for remote sensing detection tasks. Specifically, KD and AT are designed to extract prediction classification and feature attention knowledge, respectively. FGFI proposes that features near object regions are more critical, thus introducing a mask-based distillation method. FRS employs the predicted classification scores from the teacher as masks for feature-based distillation. FKD incorporates joint spatial and channel attention masks from both the teacher and the student for more effective distillation. Defeat assigns different weights to foreground and background pixels, emphasizing their equal importance in distillation. FGD employs attention masks to guide the student's focus on critical regions in the foreground and background. ARSD proposes binarized masks strategy for different object sizes. InsDist introduces novel instance-aware imitation masks. AFD combines spatial and channel attention to form advanced feature masks. Two-way designs a two-way assistant module to enhance the semantic richness of student features while reducing the complexity of teacher features. Note that the above methods are all implemented with official codes, except for those marked with the $*$ in the table, AFD [58] and two-way [27], which we reproduce based on the their original study.

We initialize all backbone networks with ImageNet pretrained weights. All students are trained using stochastic gradient descent for 12 epochs, starting with an initial learning rate of 0.005, and reducing the learning rate by 10 at the 8th and 11th epochs. We set the weight decay to 0.0001 and the momentum to 0.9. All teachers are trained for 24 epochs, starting with an initial learning rate of 0.005, and subsequently reducing the learning rate by 10 at the 16th and 22th epochs. Data augmentation techniques, involving image resizing and random flipping, are applied to both the teacher and the student during training. We use the hyperparameter settings $\alpha = 0.5$, $\beta = 7 \times 10^{-5}$, $\gamma = 0.1$, and $\lambda = 1.0$ for loss balancing. We use the hyperparameter settings $u = 0.7$, $g = 0.6$, and $t = 0.5$ for instance mining. All experiments are implemented using PyTorch [73] with the MMD framework [74], conducted on two RTX 3090 GPUs.

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D. Comparison With State-of-the-Art Methods

We present the comparative experimental results of our method and other distillation methods on the DOTA, DIOR, and SSDD datasets, as shown in Tables II–IV, respectively. In Tables II–IV, the backbone networks for the teacher and the student model are ResNet101 and ResNet18, respectively. In addition, under the Faster RCNN framework, Swin-Base, and Swin-Tiny are used as the backbone networks for the teacher and the student, respectively. Notably, the FGFI method is exclusively compatible with anchor-based detectors and thus, does not yield results for anchor-free detectors. The main findings from the results are given as follows.

1) Our DPGD consistently and effectively improves the AP across various architecture detectors on different datasets. For the DOTA dataset, there are 4.0, 3.3, and 2.4 AP improvements on anchor-based two-stage, one-stage, and anchor-free detectors, respectively. For the DIOR dataset, these improvements are even more remarkable, with increases of 5.6, 8.0, and 6.8 AP, respectively. This substantial improvement on the DIOR dataset can be attributed to the relatively powerful teacher model, which provides richer information and knowledge. For the SSDD dataset, there are 3.9, 4.1, and 4.5 AP improvements on anchor-based two-stage, one-stage, and anchor-free detectors, respectively. The outstanding results achieved on the SAR dataset show the excellent generalization capability of the proposed method.
2) Our DPGD achieves notable AP improvements of 1.4, 4.8, and 2.8 on these three datasets using Faster R-CNN with a Swin-T backbone, demonstrating its satisfactory performance on both convolutional networks and transformers.
3) Compared to 11 other distillation methods, our DPGD surpasses the suboptimal distillation method by an average of 0.7 AP on the DOTA dataset, 1.9 AP on the DIOR dataset, and 1.4 AP on the SSDD dataset. Notably, traditional KD marginally outperforms the feature-based AT method, indirectly indicating that prediction-based distillation aids students in learning task-relevant information in remote sensing detection tasks. However, due to its disregard for position information, the KD method significantly underperforms compared to mainstream mask-based feature methods.
4) We visualize the FPS-AP of detectors on the DOTA and SSDD datasets in Fig. 8. For the DOTA dataset, RetinaNet trained with our DPGD achieves a 0.3 AP improvement, slight acceleration, and 8.29 M fewer parameters compared to Faster RCNN (ResNet18 backbone) trained without knowledge distillation. For the SSDD dataset, Faster RCNN (ResNet18 backbone) trained with our DPGD achieves a 2.3 AP improvement, 1.4× acceleration, and 16.63 M fewer parameters compared to Faster RCNN (Swin-T backbone) trained without knowledge distillation. These observations suggest that the proposed method significantly accelerates and compresses detectors while maintaining or even enhancing AP.

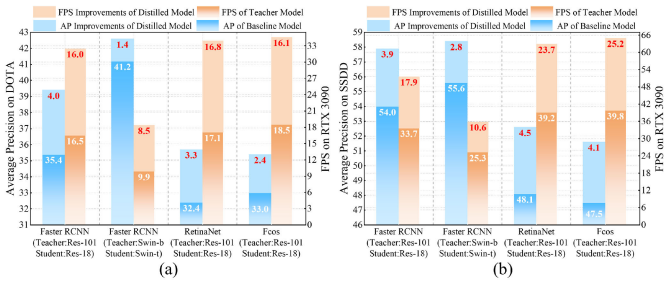


Fig. 8. FPS-AP of students with our method. (a) and (b) AP improvements for students and FPS improvements relative to teachers with our method on DOTA and SSDD datasets, respectively.

E. Ablation Study

1) Ablation Study on Each Component of Distillation: The Table V presents the results of ablation experiments on the proposed PGRD ($\mathcal{L}_{\text{ins}}$ and $\mathcal{L}_{\text{pr}}$) and PGOD ($\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{loc}}$). The ablation study is performed on the DOTA dataset using Faster RCNN with a ResNet18 backbone. The observations are as follows.

1) PGRD ($\mathcal{L}_{\text{ins}} + \mathcal{L}_{\text{pr}}$) and PGOD ($\mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{loc}}$) achieve 3.0 and 3.2 AP improvements.
2) $\mathcal{L}_{\text{ins}}$ and $\mathcal{L}_{\text{pr}}$ are capable of leading to 2.6 and 2.2 AP improvements, respectively, revealing that the majority of the advantages of PGRD are derived from instance-wise relation distillation loss.
3) $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{loc}}$ are capable of leading to 2.5 and 2.1 AP improvements, respectively, revealing that the majority of the advantages of PGOD are derived from the classification distillation loss. Additionally, prediction-based distillation performs better than feature-based distillation (see Table II). This supports our hypothesis that feature-based distillation may introduce additional noise, leading to suboptimal performance in complex remote sensing detection tasks.
4) Combining PGRD and PGOD results in an improvement of 3.0 AP. These findings emphasize the significance of each component of our DPGD, and their combined use yields enhanced performance.

2) Ablation Study on Mask Types: Unlike previous methods that use feature attention masks and static teacher-guided masks, our DPGD employs the discrepancy between teacher and student classification scores as a dynamic mask during the PGOD process. We conducted ablation studies to evaluate the impact of the three masks using the Faster RCNN (ResNet18 backbone) on the DOTA. Note that the baseline model is trained using output distillation without any type of mask. The results, presented in Table VI, show that the attention-guided mask, teacher-guided mask, and joint prediction-guided mask result in improvements of 0.4, 0.7, and 1.1 in AP, respectively. This finding demonstrates that the proposed prediction mask can dynamically adapt to the various training phases of students, leading to enhanced performance.

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TABLE II

COMPARISON BETWEEN OUR METHOD AND OTHER DISTILLATION METHODS ON THE DOTA DATASET

Model	Faster RCNN						RetinaNet						FCOS						Faster RCNN (Swin)					
	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Teacher	41.9	69.5	47.1	20.9	44.5	53.7	37.1	64.3	39.9	15.9	39.0	47.5	37.2	64.1	40.5	17.3	38.6	47.7	43.2	68.2	46.3	19.7	44.2	55.3
Student	35.4	63.4	38.0	16.5	38.3	44.1	32.4	58.0	34.2	12.6	34.2	42.1	33.0	59.2	35.7	15.2	34.9	43.1	41.2	66.7	44.2	19.3	42.6	52.5
KD [17]	36.9	64.0	40.8	17.5	39.4	47.5	32.8	58.1	35.6	13.4	33.7	43.9	33.4	59.8	36.0	14.5	34.5	43.2	41.6	66.7	44.8	19.7	43.3	52.2
AT [71]	36.7	63.8	40.5	17.4	39.2	47.6	32.6	59.1	34.8	14.1	34.3	43.0	33.2	59.5	35.6	15.3	35.2	42.1	41.3	66.7	44.4	19.3	43.2	51.9
FGFI [21]	37.7	65.1	41.7	17.8	39.9	48.8	34.1	60.1	36.7	14.1	35.0	44.4	–	–	–	–	–	–	41.7	67.4	44.8	19.1	43.3	52.9
FKD [55]	37.5	65.3	41.4	17.9	39.8	48.4	34.5	60.7	36.8	14.2	35.5	44.6	34.6	61.3	37.4	15.1	35.4	44.4	41.5	66.9	44.9	19.8	43.0	53.2
FRS [64]	38.2	65.3	42.4	18.5	40.6	49.1	35.0	61.2	37.7	14.6	35.8	46.5	34.3	60.7	37.2	14.9	35.6	44.3	42.0	67.4	45.8	19.8	43.1	53.4
Defeat [56]	37.9	65.7	41.9	18.8	40.0	49.6	34.4	61.0	37.2	14.6	35.4	44.9	34.4	60.8	37.6	15.1	35.5	44.7	41.4	66.8	44.9	18.8	42.7	52.9
FGD [57]	37.9	65.1	42.0	18.6	39.4	50.1	33.6	59.6	36.1	14.4	35.5	42.1	34.6	61.4	37.4	15.1	35.4	45.1	41.5	66.9	44.7	18.8	42.7	53.2
ARSD [26]	37.5	65.2	41.1	18.6	39.9	48.3	34.6	61.3	36.5	14.7	35.6	45.0	34.2	60.8	37.0	14.9	35.3	44.7	41.8	66.7	45.2	19.2	43.1	53.2
InsDist [24]	38.1	65.9	41.8	18.3	40.3	50.0	35.1	61.6	38.3	15.1	35.8	47.2	34.7	61.5	37.1	15.2	35.8	44.5	42.1	67.4	45.8	19.4	43.6	53.9
AFD* [58]	36.8	64.4	40.5	17.2	39.3	47.5	33.6	60.1	35.8	14.4	35.1	43.4	33.6	59.9	36.1	15.3	35.1	43.2	41.2	65.8	45.2	18.9	42.9	51.8
Two-way* [27]	36.9	64.0	40.8	17.5	39.4	47.5	33.5	59.5	35.8	14.3	34.6	42.5	33.7	60.3	36.1	14.4	34.9	43.0	41.3	65.7	45.1	19.0	42.9	51.7
Our method	**39.4**	**67.1**	**44.0**	**18.8**	**40.9**	**52.1**	**35.7**	**62.4**	**38.7**	**15.2**	**36.3**	**47.7**	**35.4**	**62.0**	**38.3**	**15.7**	**36.1**	**46.5**	**42.6**	**68.0**	**45.9**	**19.8**	**44.5**	**53.9**

TABLE III

COMPARISON BETWEEN OUR METHOD AND OTHER DISTILLATION METHODS ON THE DIOR DATASET

Model	Faster RCNN						RetinaNet						FCOS						Faster RCNN (Swin)					
	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Teacher	49.8	77.0	54.6	17.3	42.5	67.6	45.9	70.2	49.5	12.9	37.2	62.9	46.6	71.3	49.8	13.6	38.9	65.1	54.6	78.3	60.7	18.5	45.7	73.8
Student	41.3	69.4	43.0	13.2	34.3	57.1	37.1	60.2	39.4	8.4	29.7	52.7	38.6	64.2	39.6	9.1	30.4	55.4	49.1	75.3	54.1	15.0	38.3	66.5
KD [17]	43.4	70.7	46.3	13.6	35.2	59.7	38.7	61.7	42.0	8.7	29.2	54.8	41.2	65.8	43.4	10.3	33.1	58.0	50.3	75.9	55.6	14.0	39.8	68.8
AT [71]	43.1	70.6	46.1	14.1	34.9	59.4	38.3	61.6	41.7	8.6	30.1	54.8	40.8	65.7	43.3	10.4	32.8	57.4	49.8	77.0	54.5	16.9	41.8	66.1
FGFI [21]	43.8	71.2	47.1	14.4	35.8	59.9	41.4	64.2	45.1	9.6	31.5	58.8	–	–	–	–	–	–	51.3	77.8	56.5	17.6	43.4	68.3
FKD [55]	44.2	71.7	47.7	14.1	35.9	60.4	41.7	65.9	45.5	10.6	33.6	58.1	42.2	67.4	44.6	10.3	33.3	59.8	51.0	76.8	57.2	15.1	40.3	70.0
FRS [64]	44.9	71.3	48.9	13.9	34.9	62.5	42.4	66.0	45.9	10.5	33.5	58.7	42.4	67.6	45.1	10.8	33.2	59.9	51.6	76.6	57.2	15.9	41.9	69.5
Defeat [56]	44.4	71.6	47.9	14.2	36.0	60.7	41.5	64.6	45.0	10.2	32.9	58.1	42.9	68.1	45.8	11.0	33.5	60.5	51.8	78.1	57.0	17.4	43.5	69.7
FGD [57]	44.0	71.4	47.1	12.9	33.8	61.0	41.1	64.2	44.4	9.7	32.0	58.2	42.5	68.1	45.2	10.7	33.6	60.2	51.4	77.3	56,7	16.6	43.3	69.2
ARSD [26]	44.5	71.9	47.8	12.9	34.7	61.9	42.1	66.4	45.8	10.3	34.0	58.3	42.8	68.1	45.5	10.8	33.5	60.4	51.1	76.2	56.7	15.0	40.0	69.6
InsDist [24]	45.1	72.4	48.5	12.8	34.9	62.7	43.0	66.7	46.4	10.9	34.0	59.5	43.0	68.2	46.1	11.4	33.7	60.1	51.9	78.0	57.3	16.9	44.0	70.6
AFD* [58]	43.6	71.4	46.5	12.8	34.4	60.3	39.7	63.0	42.9	10.3	30.5	55.9	41.9	66.7	44.2	11.0	33.4	58.6	50.5	75.9	55.7	14.5	39.7	69.1
Two-way* [27]	43.9	71.2	47.5	12.2	33.9	61.2	39.3	63.0	42.8	10.1	31.3	55.0	42.1	66.8	44.5	11.1	33.3	58.9	50.7	76.2	56.0	14.7	40.1	69.4
Our method	**46.9**	**73.7**	**51.0**	**14.4**	**36.6**	**64.7**	**45.1**	**69.5**	**48.7**	**12.3**	**36.5**	**62.0**	**44.8**	**70.3**	**47.5**	**12.4**	**35.6**	**62.1**	**53.9**	**78.6**	**60.0**	**18.0**	**45.1**	**72.2**

TABLE IV

COMPARISON BETWEEN OUR METHOD AND OTHER DISTILLATION METHODS ON THE SSDD DATASET

Model	Faster RCNN						RetinaNet						FCOS						Faster RCNN (Swin)					
	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Teacher	59.0	93.8	66.4	53.9	67.4	64.9	52.1	88.1	56.4	47.7	60.0	50.2	53.0	87.8	57.4	48.5	60.9	53.0	59.5	94.0	68.5	54.1	68.1	69.2
Student	54.0	92.1	57.6	50.9	60.8	38.0	47.5	86.4	49.6	43.5	54.5	46.5	48.1	85.4	52.2	45.9	54.2	38.3	55.6	93.2	62.5	52.2	62.5	57.1
KD [17]	54.7	92.4	59.4	52.1	60.0	49.1	47.9	87.6	47.7	44.2	55.1	39.5	49.0	86.2	53.7	46.3	55.3	46.2	56.2	92.4	65.3	51.5	63.8	60.6
AT [71]	54.4	93.0	58.9	50.7	62.0	44.7	47.7	87.5	48.8	45.3	53.7	33.5	48.4	85.7	52.0	45.8	54.6	39.6	55.9	92.3	64.9	51.0	63.8	62.1
FGFI [21]	55.0	93.1	60.0	51.9	60.7	51.1	48.3	87.2	48.3	44.1	56.2	43.9	–	–	–	–	–	–	56.6	93.2	65.6	52.3	64.0	62.0
FKD [55]	55.1	93.1	60.1	52.0	60.9	52.4	48.5	87.1	47.9	45.4	54.7	41.1	50.0	86.8	54.1	45.9	57.7	50.9	56.7	93.1	66.0	52.1	64.3	63.1
FRS [64]	56.0	92.6	60.9	51.8	63.4	55.7	48.7	87.5	50.8	46.5	53.8	39.3	50.7	86.6	53.4	48.0	56.5	40.6	57.1	93.3	66.0	52.4	64.8	64.1
Defeat [56]	55.8	92.6	60.8	52.0	62.4	53.8	49.4	87.1	54.1	46.4	55.5	45.2	49.9	87.4	53.2	47.8	55.3	47.5	56.9	93.3	66.2	52.6	64.5	63.0
FGD [57]	55.6	93.4	60.9	51.6	62.7	57.5	48.8	87.4	50.2	46.0	54.9	40.2	50.2	86.6	54.1	47.5	55.2	41.3	56.4	94.2	61.8	52.4	64.2	63.7
ARSD [26]	55.4	93.2	60.7	51.7	62.3	57.0	49.1	86.5	52.8	47.2	54.2	36.2	50.5	86.6	54.6	47.9	56.8	39.6	56.8	92.8	64.0	52.3	64.8	60.6
InsDist [24]	56.2	92.7	62.7	51.2	63.0	55.2	50.4	88.4	55.0	47.8	56.0	47.2	51.0	87.0	54.0	47.0	58.7	52.4	57.3	93.4	65.9	52.3	65.4	66.2
AFD* [58]	54.9	92.9	61.4	52.0	60.7	52.4	50.0	88.3	55.0	46.9	56.1	45.1	49.6	86.5	51.9	47.1	55.6	45.6	56.1	93.1	62.1	52.0	63.6	63.3
Two-way* [27]	54.8	93.2	60.1	51.0	61.5	57.3	49.8	87.1	52.2	46.9	55.8	41.1	49.2	86.4	52.2	46.6	55.1	42.4	55.8	92.8	61.3	52.1	62.5	56.7
Our method	**57.9**	**93.6**	**64.8**	**52.3**	**67.3**	**62.7**	**51.6**	**88.7**	**57.6**	**48.3**	**58.1**	**50.5**	**52.6**	**87.9**	**57.4**	**48.1**	**60.4**	**53.4**	**58.4**	**93.7**	**68.2**	**53.3**	**66.6**	**67.6**

TABLE V

ABLATION STUDY OF THE FOUR DISTILLATION LOSSES IN OUR METHOD

Loss	$\mathcal{L}_{\text{ins}}$	×	✓	×	×	×	✓	✓	✓	×	×	×	✓	✓	✓	×	✓
	$\mathcal{L}_{\text{pr}}$	×	×	✓	×	×	✓	×	×	✓	✓	×	✓	✓	×	✓	✓
	$\mathcal{L}_{\text{cls}}$	×	×	×	✓	×	×	✓	×	✓	×	✓	✓	×	✓	✓	✓
	$\mathcal{L}_{\text{loc}}$	×	×	×	×	✓	×	×	✓	×	✓	✓	×	✓	✓	✓	✓
Result	**AP**	35.4	38.0	37.6	37.9	37.5	38.5	38.7	38.3	38.4	38.1	38.6	39.1	38.9	39.0	38.7	**39.4**
	$\mathbf{AP}_S$	16.5	18.1	18.5	17.7	18.0	18.5	18.8	18.8	18.1	18.3	18.8	18.8	18.6	18.6	18.7	**18.8**
	$\mathbf{AP}_M$	38.3	40.1	39.5	40.0	39.8	40.8	40.6	40.6	40.7	40.0	40.6	40.8	40.2	40.5	40.7	**40.9**
	$\mathbf{AP}_L$	44.1	49.8	49.0	49.4	49.5	50.0	51.2	49.7	49.5	50.7	51.3	52.0	51.6	51.9	51.2	**52.1**

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TABLE VI
ABLATION STUDY OF THE THREE MASKS IN OUR METHOD

Mask Type					
	Attention-Guided	×	✓	×	×
	Teacher-Guide	×	×	✓	×
	Joint Prediction-Guided	×	×	×	✓
Result	AP	37.5	37.9	38.2	**38.6**
	AP_S	17.7	17.7	18.3	**18.8**
	AP_M	39.8	39.9	40.6	**40.6**
	AP_L	49.0	49.8	49.6	**51.3**

TABLE VII
HYPERPARAMETER SENSITIVITY STUDY OF u, g, AND t USED FOR INSTANCE MINING WITH FASTER RCNN WITH RESNET18 BACKBONE ON THE DOTA DATASET

Value		0.0	0.1	0.2	0.3	0.4	**0.5**	**0.6**	**0.7**	0.8	0.9	1.0
u	AP	35.5	37.3	37.3	37.4	37.7	37.8	37.9	**38.0**	37.8	37.6	37.7
	AP_S	17.3	18.2	17.6	18.0	**18.9**	17.8	17.6	18.1	18.4	18.1	17.8
	AP_M	37.6	39.8	40.1	40.2	39.8	39.8	**40.6**	40.1	40.1	39.9	39.8
	AP_L	44.6	48.8	49.1	49.1	48.1	49.9	49.7	**49.9**	48.6	48.2	49.8
g	AP	37.6	37.6	37.6	37.7	37.8	37.8	**38.0**	37.9	37.9	37.5	37.5
	AP_S	18.0	**18.2**	18.0	17.7	17.8	**18.2**	18.1	**18.2**	17.7	18.1	18.1
	AP_M	40.0	40.3	39.7	39.9	40.0	**40.6**	40.1	40.5	40.0	40.1	39.7
	AP_L	49.7	48.6	49.3	49.6	49.7	48.0	**49.9**	48.6	49.4	48.6	49.4
t	AP	37.6	37.6	37.7	37.8	37.9	**38.0**	37.9	37.8	37.9	37.6	37.6
	AP_S	18.0	18.3	17,7	18.3	19.2	18.1	18.1	**18.6**	18.2	18.5	18.1
	AP_M	40.0	39.9	**40.4**	40.3	40.3	40.1	**40.4**	39.7	40.0	39.1	39.8
	AP_L	49.6	**49.9**	48.8	49.0	48.9	**49.9**	48.8	48.7	48.8	48.9	48.4

F. Sensitivity Study

1) Sensitivity Study on Hyperparameters of Distillation Loss: We investigate the sensitivity of four hyperparameters in the proposed distillation loss: α, β, γ, and λ. These hyperparameters balance the magnitudes of different types of distillation losses. The sensitivity study is performed on the DOTA dataset using Faster RCNN with a ResNet18 backbone. As shown in Fig. 9, even the poorest hyperparameter selected, there is only a 0.3 AP decrease compared to the optimal setting, while still achieving a 3.7 AP improvement over the baseline student model. This indicates that our DPGD is robust to hyperparameter choices.

2) Sensitivity Study on Hyperparameters of Instance Selection: We investigate the sensitivity of three hyperparameters in the instance-wise relation distillation: u, g, and t, as shown in Table VII. These correspond to the threshold for the NMS operation, the threshold for IoU computation, and the threshold for classification loss in instance selection, respectively. It can be observed that as follows.

1) The NMS operation improves clarity and reliability by removing highly overlapping candidate boxes, resulting in performance gains. When $u = 0$, only the highest-scoring candidate box is retained, providing minimal benefits. Conversely, when $u = 1$, the NMS is not employed, leading to excessive candidate boxes and unnecessary computational overhead.
2) The IoU computation aids in distinguishing between positive and negative instances. When $g = 1$, there is a 0.5 drop in AP, indicating that focusing solely on features of ground truth neglects important background details.
3) Classification loss aids in selecting valuable background instances. When $t = 0$ or $t = 1$, there are drops of 0.2 and 0.3 in AP, respectively. This highlights the significance of background in the instance-wise relation distillation process, although not all background instances are equally crucial. These findings demonstrate that our joint prediction-guided selection method effectively chooses instances in relation distillation and robustly selects hyperparameters. These three hyperparameters u, g, and t are similar to other parameters in the object detection training process. For example, in the Faster RCNN framework, the threshold for NMS filtering is typically set to 0.7, and the IoU threshold for selecting positive samples is also 0.7, which is very close to the optimal value we used. Therefore, u and g can be set based on the corresponding parameters used in training the detection framework. In practice, only the t in the instance mining module needs to be manually tuned.

G. Visualization Analysis

1) Qualitative Analysis of Benefits of Knowledge Distillation: Fig. 10 shows the detection results of the detector trained with and without the proposed distillation method. The findings are presented as follows.

1) DPGD improves the detection capability of student model to detect small objects. Fig. 10(a)–(d) shows that the distilled model is able to detect significantly more small vehicles, planes, and ships.
2) DPGD enhances the detection capability for dense remote sensing objects. In Fig. 10(b), (c), and (e), compared to the original student model, the distilled model identifies remarkably more planes, ships, and small vehicles.
3) DPGD improves the quality of the predicted bounding boxes. Fig. 10(f)–(h) shows that the original student model produces multiple bounding boxes for harbors and ground-track fields, whereas the distilled model avoids such errors.
4) DPGD delivers more accurate classification scores for objects. In Fig. 10(i) and (j), the original model detects ships with low confidence and misses similar ones. Still, the distilled model enhances both the confidence and accuracy of the detection results. These findings indicate that DPGD positively impacts different facets of detection results, effectively correcting errors in the original model.

2) Visualization of Training Process: Fig. 11 illustrates the training process of Faster RCNN with a ResNet18 on DOTA, focusing on the individual distillation losses and AP. The findings are presented as follows.

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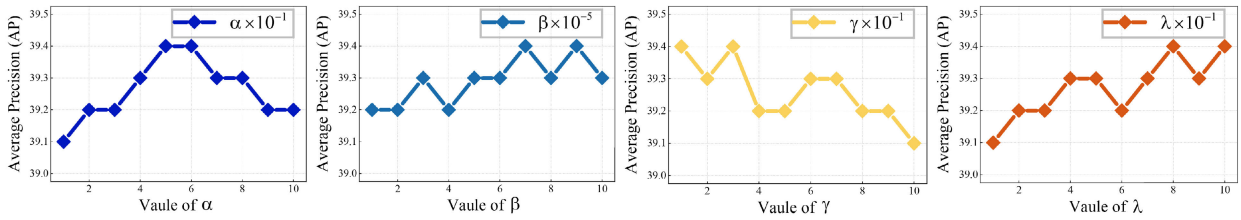


Fig. 9. Hyperparameter sensitivity study of α, β, γ, and λ used for loss balancing with Faster RCNN with ResNet18 backbone on the DOTA dataset.

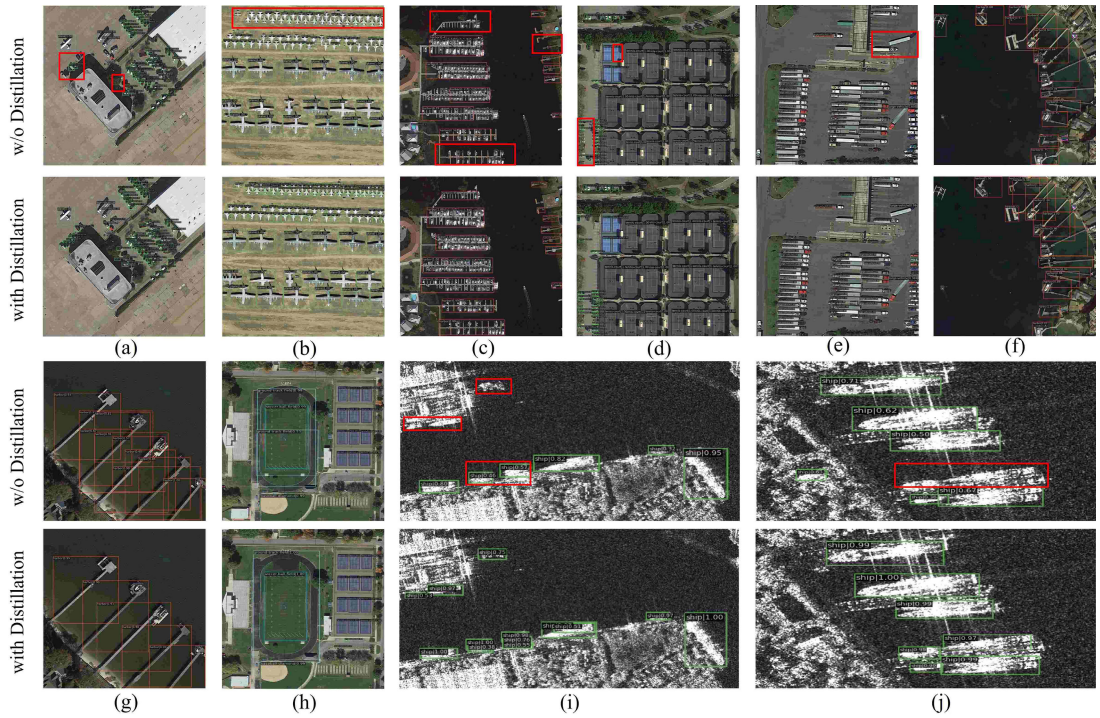


Fig. 10. Qualitative analysis with Faster RCNN (ResNet18 backbone) with and without our method training. (a)–(h) Display the detection results of eight different input images on the DOTA. (i) and (j) Detection results of two different input images on the SSDD.

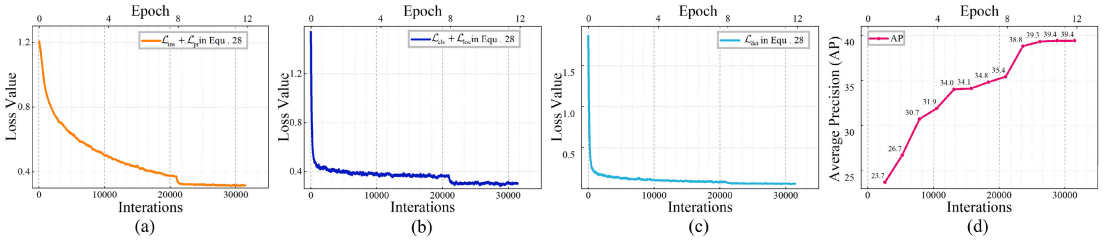


Fig. 11. Curves of different loss and AP during distilled Faster RCNN training using a ResNet18 backbone on the DOTA dataset. (a) and (b) Loss curves of the proposed relation and output distillation, respectively. (c) Training loss of the original detection task for the student model. Notably, at the eighth epoch, both the AP improves and the loss decreases significantly due to the learning rate being reduced at this point. (d) Demonstrates the AP trained with our method on the validation set for different training epochs.

1) In the initial iterations (0k–5k), both the relation and output distillation losses, as well as the original task loss, exhibit a rapid decrease.
2) In the following iterations (5k–21k), the primal detection task loss tends to stabilize, while the distillation losses continue to decrease significantly. Therefore, the student detector with the proposed DPGD method performs better than without it. This indicated that distillation losses provide additional supervisory signals from the teacher model to further optimize the student model, especially when the original task loss provides not enough information.
3) In the last iterations, following the epoch of the learning rate decayed, detectors with and without distillation training show performance improvements, while the distillation-enhanced student yielding superior results. These findings indicate that distillation losses are consistently reduced throughout the training process, implying

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that they pose a more noticeable optimization challenge than the original task loss.

V. Conclusion and Future Work

In this study, we propose DPGD methods, consisting of prediction-guided relational distillation and PGOD, for object detection tasks in remote sensing. Our prediction-guided instance-wise relation distillation initially identifies candidate instances based on the joint predictions from both teacher and student models. To select valuable instances, these candidates are refined through NMS, IoU calculation, and classification loss. This enables the student to concentrate on global semantic instance relations rather than relying solely on the local relations guided by ground truth. In addition, our prediction-guided pixel-wise relation distillation employs nonlocal modules to capture fine-grained relations among pixels. Simultaneously, our PGOD uses distillation losses based on binary cross entropy and GIoU, which are consistent and compatible with the original task to transfer classification and localization information, respectively. This enables the student to learn essential knowledge directly related to detection tasks. We test 11 distillation methods and four types of detectors to emphasize the effective performance of our method on anchor-based two-stage, one-stage, and no-anchor models over three remote sensing detection datasets. We conduct comprehensive ablation studies for various distillation losses and mask types to demonstrate the essential role of each component in proposed method. Furthermore, we study the effect of hyperparameters for balancing different distillation losses and instance selection to illustrate the robustness of our method to hyperparameter choices. To further explore the benefits of distillation learning, we provide visualizations of the detection results and the distillation losses, showing that the extra supervision information from the teacher improves the classification and localization abilities of the detectors.

In future work, we plan to integrate knowledge distillation with additional model compression techniques, like pruning and quantization, to achieve superior performance and faster inference speeds. In addition, we are interested in exploring a unified knowledge distillation framework for all remote sensing image tasks, such as classification and semantic segmentation.

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